

Ostbayerische Technische Hochschule Regensburg

Studiengang Informatik

Programmieren 2 SS 2026

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Übungsblatt 6 - Lösung

Aufgabe 1

```
public class Addition implements Function {  
  
    public double getResult(double x, double y) {  
        return x + y;  
    }  
  
    public String toString() {  
        return "+";  
    }  
}  
  
public class Multiplication implements Function {  
  
    public double getResult(double x, double y) {  
        return x * y;  
    }  
  
    public String toString() {  
        return "*";  
    }  
}
```

Aufgabe 2

```
public class Calculator {  
  
    private Stack<Double> history;  
  
    public Calculator() {  
        history = new Stack<Double>();  
        clear();  
    }  
  
    public void clear() {  
        history.clear();  
        history.push(0.0);  
    }  
  
    public void apply(Function f, double y) {  
        Double r = f.getResult(getResult(), y);  
        history.push(r);  
    }  
}
```

```

    public void undo() {
        if (canUndo()) history.pop();
    }

    public boolean canUndo() {
        return history.size() > 1;
    }

    public double getResult() {
        return history.peek();
    }
}

```

Aufgabe 3

```

public class CalculatorFrame extends JFrame {

    class CalcActionHandler implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            try {
                double argument =
                    Double.parseDouble(txtArgument.getText());
                Function f =
                    (Function) cboFunction.getSelectedItem();
                calculator.apply(f, argument);
                updateState();
            }
            catch (NumberFormatException ex) {
                txtResult.setText("Ungültige Eingabe!");
            }
        }
    }

    class UndoActionHandler implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            calculator.undo();
            updateState();
        }
    }

    class ClearActionHandler implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            calculator.clear();
            updateState();
        }
    }

    private Calculator calculator;

    private JMenuItem miUndo;
    private JButton cmdUndo;

    private JTextField txtResult;
    private JComboBox cboFunction;
    private JTextField txtArgument;

    public CalculatorFrame() {

        calculator = new Calculator();
    }
}

```

```

setSize(300,80);
setDefaultCloseOperation(DISPOSE_ON_CLOSE);

JMenuBar menubar = new JMenuBar();
setJMenuBar(menubar);

JMenu menu;
JMenuItem mi;

menubar.add(menu = new JMenu("Datei"));

menu.add(mi = new JMenuItem("Beenden"));
mi.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {
        dispose();
    }
});

menubar.add(menu = new JMenu("Bearbeiten"));

menu.add(mi = new JMenuItem("Berechnen"));
mi.addActionListener(new CalcActionHandler());

menu.add(miUndo = new JMenuItem("Rückgängig"));
miUndo.addActionListener(new UndoActionHandler());

menu.add(mi = new JMenuItem("Zurücksetzen"));
mi.addActionListener(new ClearActionHandler());


JPanel panel;

panel = new JPanel(new GridLayout(3, 2));
add(panel);

panel.add(new JLabel("Ergebnis:"));
panel.add(txtResult = new JTextField());
txtResult.setEditable(false);

panel.add(new JLabel("Funktion:"));
panel.add(cboFunction = new JComboBox());
cboFunction.addItem(new Addition());
cboFunction.addItem(new Multiplication());

panel.add(new JLabel("Argument:"));
panel.add(txtArgument = new JTextField());

panel = new JPanel();
add(panel, BorderLayout.SOUTH);

JButton cmd;

panel.add(cmd = new JButton("Berechnen"));
cmd.addActionListener(new CalcActionHandler());
getRootPane().setDefaultButton(cmd);

panel.add(cmdUndo = new JButton("Rückgängig"));
cmdUndo.addActionListener(new UndoActionHandler());

panel.add(cmd = new JButton("Zurücksetzen"));
cmd.addActionListener(new ClearActionHandler());

```

```
        updateState();

        pack();
        setVisible(true);

        cboFunction.requestFocusInWindow();
    }

    private void updateState() {
        Double r = calculator.getResult();
        txtResult.setText(r.toString());

        miUndo.setEnabled(calculator.canUndo());
        cmdUndo.setEnabled(calculator.canUndo());
    }

    public static void main(String[] args) {
        new CalculatorFrame();
    }
}
```